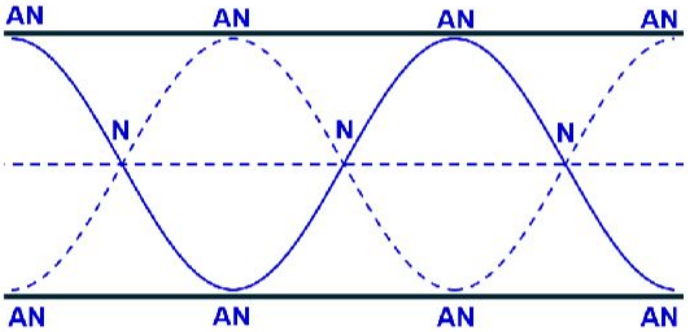
Qns	Answer	Marks
6(a)	The principle of superposition states that when two or more waves meet at a point, the resultant displacement is the sum of the individual displacements of each wave at that point. **Note:** Do **NOT** accept if students mention the "amplitude add up". The correct term is displacement.	**B1**
6(b)	Incident and reflected waves are of the same type, frequency and wavelength, and have the same speed, approximately the same amplitude and travel in opposite direction along the same tube. Incident wave from the loudspeaker reflects at the end / right side of the glass tube such that they superpose to form a stationary wave. **Note:** Do not accept if students write meet / overlap without superpose. The word interfere is not accepted as well.	**B1** **B1** **B1**
6(c)(i)	Amplitude of stationary wave: 0.06 + 0.06 = 0.12 mm	**B1**

Qns	Answer	Marks
6(c)(ii)	 General shape (include spacing) of a stationary wave (including dotted cosine curve – no solid line) with displacement antinodes at the ends of the glass tube. Stationary waves fulfilling the 3rd harmonic, n = 3, since there are three pressure antinodes (three displacement nodes) *#Although labelling **N** and **AN** is not a marking point, it is good practice to write it down.* **Note:** Students should know that a pressure antinode corresponds to a displacement node. Hence, there should be 3 displacement nodes which correspond to the 3rd harmonic for a pipe with two open ends.	**B1** **B1**
6(c)(iii)	$L = 1.5\lambda \Rightarrow \lambda = \dfrac{L}{1.5}$ $v = f\lambda \Rightarrow f = \dfrac{v}{\lambda} = \dfrac{v}{\left(\frac{L}{1.5}\right)} = \dfrac{320}{\left(\frac{15.0}{1.5}\right)} = 32$ Hz	**M0** **A1**
7(a)(i)	${}^{210}_{83}Bi \rightarrow {}^{0}_{-1}e + {}^{210}_{84}Po$ + (anti) neutrino	**B2**